Course- B.Tech.
Course Code- CSET343
Year- 4TH Year

Type- Specialization Elective
Course Name- AI in Healthcare
Semester- VII

Lab Assignment-10

Objective:

To build a complete beat-level arrhythmia classifier on the MIT-BIH Arrhythmia Database and compare three recurrent architectures — RNN (Recurrent Neural Network), LSTM (Long-Short Term Memory) and GRU (Gated Recurrent Unit)

Dataset:

MIT-BIH Arrhythmia Dataset

The MIT-BIH Arrhythmia Database contains 48 half-hour excerpts of two-channel ambulatory ECG recordings, obtained from 47 subjects studied by the BIH Arrhythmia Laboratory between 1975 and 1979.

Task 1: Data Acquisition and Exploration

- **Data Acquisition:** Download the MIT-BIH database from PhysioNet (48 two-lead recordings, 360 Hz) using wfdb.
- **Signal Inspection & Cleaning:** Plot signals, check class distribution (AAMI classes N, S, V, F, Q), and remove paced records and non-beat annotations.

Task 2: Data Preprocessing

- **Noise Removal / Filtering:** Remove baseline wander and noise with a 0.5–40 Hz bandpass or wavelet denoising.
- **Segmentation:** Extract fixed-length windows around each R-peak with its beat label.
- **Feature Extraction:** Use raw beat morphology, optionally with RR intervals or wavelet features.
- **Normalization:** Normalize each beat window (and/or extracted feature vector) to a consistent scale
- **Data Split:** Create training, validation and test sets (preferably inter-patient) and handle class imbalance.

Task 3: Model Training and Evaluation

- **Model Design:** Build comparable RNN, LSTM and GRU classifiers.
- **Training and Validation:** Train all three with identical settings and early stopping.
- **Testing and Performance Comparison:** Compare accuracy, precision, recall, F1 and confusion matrices on the test set.